# AQ4.2: Activity Questions 2 - Not Graded

This assignment will not be graded and is only for practice.
**Level 1**

*1 point*
Match the vector spaces (with the usual scalar multiplication and vector addition as in $M_{3\times 3}(R)$M3×3 (R) ) in column A with their bases in column B in Table : M2W4AQ2.

Choose the correct option.

- [ ] a → → ii
- [ ] a → → i
- [ ] b → → iii
- [ ] c → → i
- [ ] b → → ii
- [ ] c → → iii

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
a → → ii
b → → iii
c → → i

If $S$S be a subset of $R^5$R5 such that $span(S)$ = $R^5$span(S) = R5, then what is the minimum number of elements possible in $S$S?

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 5

*1 point*
*1 point*
If $S_1$S1 is a maximal linear independent set and $S_2$S2 is a minimal spanning set of a vector space $V$V , then which of the following option(s) is (are) true?

- [ ] For any $v \in S_1$v ∈ S1, $S_1 \setminus \{v\}$S1 \ {v} is a linearly independent.
- [ ] For any $v \in S_2$v ∈ S2, $S_2 \setminus \{v\}$S2 \ {v} is a spanning set of $V$V .
- [ ] For any $v \in V \setminus S_1$v ∈ V \ S1, $S_1 \cup \{v\}$S1 ∪ {v} is a linearly dependent.
- [ ] For any $v \in V$v ∈ V , $S_2 \cup \{v\}$S2 ∪ {v} is a spanning set of $V$V .

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**

For any $v \in S_1$ v ∈ S1, $S_1 \setminus \{v\}$S1 \ {v} is a linearly independent.

For any $v \in V \setminus S_1$ v ∈ V \ S1, $S_1 \cup \{v\}$S1 ∪ {v} is a linearly dependent.

For any $v \in V$v ∈ V , $S_2 \cup \{v\}$S2 ∪ {v} is a spanning set of $V$V .

*1 point*

Consider the following subset of $R^2$ R2 with usual addition and scalar multiplication as in $R^2$ R2.

- $V_1 = \{(x, 0) \mid x \in R\}$V1 = {(x, 0) | x ∈ R}
- $V_2 = \{(0, y) \mid y \in R\}$V2 = {(0, y) | y ∈ R}

Which of the following options are correct?

- [ ] Both $V_1$ V1 and $V_2$ V2 are subspaces of $R^2$ R2.
- [ ] $V_1 \cap V_2$ V1 ∩ V2 is a subspace of $R^2$ R2.
- [ ] $V_1 \cup V_2$ V1 ∪ V2 is a subspace of $R^2$ R2.
- [ ] $V_1$ V1 is a subspace of $R^2$ R2, but $V_2$ V2 is not.
- [ ] $V_2$ V2 is a subspace of $R^2$ R2, but $V_1$ V1 is not.
- [ ] $\{(1, 0)\}$ {(1, 0)} is a basis of $V_1$ V1.
- [ ] $\{(0, 1)\}$ {(0, 1)} is a basis of $V_2$ V2.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Both $V_1$ V1 and $V_2$ V2 are subspaces of $R^2$ R2.

$V_1 \cap V_2$ V1 ∩ V2 is a subspace of $R^2$ R2.

$\{(1, 0)\}$ {(1, 0)} is a basis of $V_1$ V1.

$\{(0, 1)\}$ {(0, 1)} is a basis of $V_2$ V2.

*1 point*

Consider the following subset of $R^2$ R2 with usual addition and scalar multiplication as in $R^2$ R2.

- $V_1 = \{(x, 0) \mid x \in R\}$V1 = {(x, 0) | x ∈ R}
- $V_2 = \{(2x, 0) \mid x \in R\}$V2 = {(2x, 0) | x ∈ R}

Which of the following options are correct?

- [ ] Both $V_1$ V1 and $V_2$ V2 are subspaces of $R^2$ R2.
- [ ] $V_1 \cap V_2$ V1 ∩ V2 is a subspace of $R^2$ R2.
- [ ] $V_1 \cup V_2$ V1 ∪ V2 is a subspace of $R^2$ R2.
- [ ] $V_1$ V1 is a subspace of $R^2$ R2, but $V_2$ V2 is not.
- [ ] $V_2$ V2 is a subspace of $R^2$ R2, but $V_1$ V1 is not.
- [ ] $\{(1, 0)\}$ {(1, 0)} is a basis of $V_1$ V1.
- [ ] $\{(1, 0)\}$ {(1, 0)} is a basis of $V_2$ V2.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**
Both $V_1$ V1 and $V_2$ V2 are subspaces of $R^2$ R2.
$V_1 \cap V_2$ V1 ∩ V2 is a subspace of $R^2$ R2.
$V_1 \cup V_2$ V1 ∪ V2 is a subspace of $R^2$ R2.
$\{(1,0)\}$ $\{(1,0)\}$ is a basis of $V_1$ V1.
$\{(1,0)\}$ $\{(1,0)\}$ is a basis of $V_2$ V2.

*1 point*
Let $V$ V be a vector space which is defined as follows:

$$V = \{(x, y, z, w) \mid x + z = y + w\} \subseteq R^4 \text{ V } = \{(\text{x}, \text{y}, \text{z}, \text{w}) \mid \text{x} + \text{z} = \text{y} + \text{w}\} \subseteq \text{R4}$$

with usual addition and scalar multiplication. Which of the following set forms a basis of $V$ V ?

- [ ] $\{(1,0,0,0),(0,1,0,0),(0,0,0,1)\}$ $\{(1,0,0,0),(0,1,0,0),(0,0,0,1)\}$.
- [ ] $\{(1,1,0,0),(0,1,-1,0),(0,-1,0,1)\}$ $\{(1,1,0,0),(0,1,-1,0),(0,-1,0,1)\}$.
- [ ] $\{(1,0,-1,0),(1,1,0,0),(1,0,0,1)\}$ $\{(1,0,-1,0),(1,1,0,0),(1,0,0,1)\}$.
- [ ] $\{(1,1,0,0),(0,1,1,0),(0,-1,0,1)\}$ $\{(1,1,0,0),(0,1,1,0),(0,-1,0,1)\}$.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$\{(1,0,-1,0),(1,1,0,0),(1,0,0,1)\}$ $\{(1,0,-1,0),(1,1,0,0),(1,0,0,1)\}$.
$\{(1,1,0,0),(0,1,1,0),(0,-1,0,1)\}$ $\{(1,1,0,0),(0,1,1,0),(0,-1,0,1)\}$.

**Level 2**

*1 point*
Which of the following sets form a basis of the vector space of $2 \times 2$ $2 \times 2$ lower triangular real matrices with usual matrix addition and scalar multiplication? (More than one option may be correct)

- [ ] $\{\begin{bmatrix}1&0\\0&0\end{bmatrix}, \begin{bmatrix}0&0\\0&1\end{bmatrix}, \begin{bmatrix}0&0\\1&0\end{bmatrix}\}$ $\{[1000], [0001], [0100]\}$
- [ ] $\{\begin{bmatrix}1&0\\1&0\end{bmatrix}, \begin{bmatrix}0&0\\1&1\end{bmatrix}, \begin{bmatrix}1&0\\0&1\end{bmatrix}\}$ $\{[1100], [0101], [1001]\}$
- [ ] $\{\begin{bmatrix}0&0\\1&0\end{bmatrix}\}$ $\{[0100]\}$
- [ ] $\{\begin{bmatrix}-1&0\\0&0\end{bmatrix}, \begin{bmatrix}0&0\\0&-1\end{bmatrix}, \begin{bmatrix}0&0\\1&0\end{bmatrix}\}$ $\{[-1000], [000-1], [0100]\}$

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$\{\begin{bmatrix}1&0\\0&0\end{bmatrix}, \begin{bmatrix}0&0\\0&1\end{bmatrix}, \begin{bmatrix}0&0\\1&0\end{bmatrix}\}$ $\{[1000], [0001], [0100]\}$
$\{\begin{bmatrix}1&0\\1&0\end{bmatrix}, \begin{bmatrix}0&0\\1&1\end{bmatrix}, \begin{bmatrix}1&0\\0&1\end{bmatrix}\}$ $\{[1100], [0101], [1001]\}$
$\{\begin{bmatrix}-1&0\\0&0\end{bmatrix}, \begin{bmatrix}0&0\\0&-1\end{bmatrix}, \begin{bmatrix}0&0\\1&0\end{bmatrix}\}$ $\{[-1000], [000-1], [0100]\}$

If $S$S is a linearly independent set in vector space $V = \{(x, y) \mid y = 2x,$ where $x, y \in R\}$V = {(x, y) | y = 2x, where x, y ∈ R} with usual addition and scalar multiplication as in $R^2$ R2, then find the maximum possible cardinality of $S$S.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 1
*1 point*
*1 point*
If $\{v_1, v_2, v_3\}$ {v1, v2, v3} forms a basis of $R^3$ R3, then which of the following are true?

- [ ] $\{v_1, v_2, v_1 + v_3\}$ {v1, v2, v1 + v3} forms a basis of $R^3$ R3
- [ ] $\{v_1, v_1 + v_2, v_1 + v_3\}$ {v1, v1 + v2, v1 + v3} forms a basis of $R^3$ R3.
- [ ] $\{v_1, v_1 + v_2, v_1 - v_3\}$ {v1, v1 + v2, v1 − v3} forms a basis of $R^3$ R3.
- [ ] $\{v_1, v_1 - v_2, v_1 - v_3\}$ {v1, v1 − v2, v1 − v3} forms a basis of $R^3$ R3.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$\{v_1, v_2, v_1 + v_3\}$ {v1, v2, v1 + v3} forms a basis of $R^3$ R3
$\{v_1, v_1 + v_2, v_1 + v_3\}$ {v1, v1 + v2, v1 + v3} forms a basis of $R^3$ R3.
$\{v_1, v_1 + v_2, v_1 - v_3\}$ {v1, v1 + v2, v1 − v3} forms a basis of $R^3$ R3.
$\{v_1, v_1 - v_2, v_1 - v_3\}$ {v1, v1 − v2, v1 − v3} forms a basis of $R^3$ R3.
*1 point*
Which of the following options is(are) true?

- [ ] Any minimal spanning set of a vector space $V$V must be a basis of $V$V .
- [ ] Any maximal spanning set of a vector space $V$V must be a basis of $V$V .
- [ ] Any minimal linear independent set of vector space $V$V must be a basis of $V$V .
- [ ] Any maximal linear independent set of vector space $V$V must be a basis of $V$V .
- [ ] The basis of a vector space is unique.
- [ ] The number of elements in a basis of $R^3$ R3 is 33.
- [ ] The number of elements in a basis of $M_{3\times 3}(R)$M3×3(R) is 33.
- [ ] There are infinite number of bases of $R^3$ R3.
- [ ] Any subset of a minimal spanning set of $V$V cannot be a spanning set.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
Any minimal spanning set of a vector space $V$V must be a basis of $V$V .
Any maximal linear independent set of vector space $V$V must be a basis of $V$V .
The number of elements in a basis of $R^3$ R3 is 33.

There are infinite number of bases of $R^3$ R3.
Any subset of a minimal spanning set of $V$ V cannot be a spanning set.

Check Answers
Your score is: 0/10